1.Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time.

a) FCFS b) SJF non preemptive c) SJF preemptive

```
#include <stdio.h>
#include <limits.h>

#define MAX 100

// Function to calculate FCFS
void fcfs(int n, int at[], int bt[]) {
   int wt[MAX], tat[MAX], ct[MAX];

   ct[0] = at[0] + bt[0];
   wt[0] = 0;
   tat[0] = bt[0];

   for(int i = 1; i < n; i++) {
      if(ct[i-1] < at[i])
         ct[i] = at[i] + bt[i];
      else
         ct[i] = ct[i-1] + bt[i];

      wt[i] = ct[i] - at[i] - bt[i];
      tat[i] = ct[i] - at[i];
   }

   float avg_wt = 0, avg_tat = 0;
```

```
    printf("\nFCFS Scheduling:\n");
    printf("PID\tAT\tBT\tWT\tTAT\n");

    for(int i = 0; i < n; i++) {
        avg_wt += wt[i];
        avg_tat += tat[i];
        printf("%d\t%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], wt[i], tat[i]);
    }

    printf("Average Waiting Time = %.2f\n", avg_wt/n);
    printf("Average Turnaround Time = %.2f\n", avg_tat/n);
}

// Function for SJF Non-Preemptive
void sjf_non_preemptive(int n, int at[], int bt[]) {
    int completed = 0, time = 0;
    int wt[MAX], tat[MAX], ct[MAX];
    int isCompleted[MAX] = {0};

    while(completed != n) {
        int idx = -1;
        int min = INT_MAX;

        for(int i = 0; i < n; i++) {
            if(at[i] <= time && !isCompleted[i]) {
                if(bt[i] < min) {
                    min = bt[i];
                    idx = i;
```

```
            }
        }
    }

    if(idx != -1) {
        time += bt[idx];
        ct[idx] = time;
        tat[idx] = ct[idx] - at[idx];
        wt[idx] = tat[idx] - bt[idx];
        isCompleted[idx] = 1;
        completed++;
    } else {
        time++;
    }
}

float avg_wt = 0, avg_tat = 0;

printf("\nSJF Non-Preemptive Scheduling:\n");
printf("PID\tAT\tBT\tWT\tTAT\n");

for(int i = 0; i < n; i++) {
    avg_wt += wt[i];
    avg_tat += tat[i];
    printf("%d\t%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], wt[i], tat[i]);
}

printf("Average Waiting Time = %.2f\n", avg_wt/n);
printf("Average Turnaround Time = %.2f\n", avg_tat/n);
```

```
}

// Function for SJF Preemptive (SRTF)
void sjf_preemptive(int n, int at[], int bt[]) {
    int rt[MAX], wt[MAX], tat[MAX];
    int completed = 0, time = 0;
    int min = INT_MAX, shortest = -1;
    int finish_time;
    int check = 0;

    for(int i = 0; i < n; i++)
        rt[i] = bt[i];

    while(completed != n) {
        for(int i = 0; i < n; i++) {
            if(at[i] <= time && rt[i] < min && rt[i] > 0) {
                min = rt[i];
                shortest = i;
                check = 1;
            }
        }

        if(!check) {
            time++;
            continue;
        }

        rt[shortest]--;
        min = rt[shortest];
```

```
    if(min == 0)
        min = INT_MAX;

    if(rt[shortest] == 0) {
        completed++;
        check = 0;
        finish_time = time + 1;

        wt[shortest] = finish_time - bt[shortest] - at[shortest];
        if(wt[shortest] < 0)
            wt[shortest] = 0;
    }
    time++;
}

float avg_wt = 0, avg_tat = 0;

printf("\nSJF Preemptive (SRTF) Scheduling:\n");
printf("PID\tAT\tBT\tWT\tTAT\n");

for(int i = 0; i < n; i++) {
    tat[i] = bt[i] + wt[i];
    avg_wt += wt[i];
    avg_tat += tat[i];
    printf("%d\t%d\t%d\t%d\t%d\n", i+1, at[i], bt[i], wt[i], tat[i]);
}

printf("Average Waiting Time = %.2f\n", avg_wt/n);
printf("Average Turnaround Time = %.2f\n", avg_tat/n);
```

```
}

int main() {
    int n;
    int at[MAX], bt[MAX];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(int i = 0; i < n; i++) {
        printf("Enter Arrival Time and Burst Time for Process %d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
    }

    fcfs(n, at, bt);
    sjf_non_preemptive(n, at, bt);
    sjf_preemptive(n, at, bt);

    return 0;
}
```

C:\Users\BMSCECSE-I4\Deskt

```
Enter number of processes: 4
Enter Arrival Time and Burst Time for Process 1: 9
5
Enter Arrival Time and Burst Time for Process 2: 7
8
Enter Arrival Time and Burst Time for Process 3: 4
3
Enter Arrival Time and Burst Time for Process 4: 0
2

FCFS Scheduling:
PID     AT      BT      WT      TAT
1       9       5       0       5
2       7       8       7       15
3       4       3       18      21
4       0       2       25      27
Average Waiting Time = 12.50
Average Turnaround Time = 17.00

SJF Non-Preemptive Scheduling:
PID     AT      BT      WT      TAT
1       9       5       6       11
2       7       8       0       8
3       4       3       0       3
4       0       2       0       2
Average Waiting Time = 1.50
Average Turnaround Time = 6.00

SJF Preemptive (SRTF) Scheduling:
PID     AT      BT      WT      TAT
1       9       5       0       5
2       7       8       5       13
3       4       3       0       3
4       0       2       0       2
Average Waiting Time = 1.25
Average Turnaround Time = 5.75

Process returned 0 (0x0)   execution time : 34.027 s
Press any key to continue.
```